EXPERIMENT-11

De-Morgan's Theorem

AIM: Implement a logic circuit to verify demorgans theorem.

COMPONENTS REQUIRED:Trainer kit

THEORY:

De Morgan's Law Boolean Algebra defines the relation between the OR, AND, and the complements of variables, and is given for both the complement of AND and OR of two values. In Boolean Algebra there are two De Morgan's Laws that are:

- First De Morgan's Law
- Second De Morgan's Law

Let's understand these laws in detail as below:

First De Morgan's Law in Boolean Algebra

First De Morgan's law states that **"The complement of OR of two or more variables is equal to the AND of the complement of each variable."**
Let A and B be two variables, then mathematically First De Morgan's Law is given as:

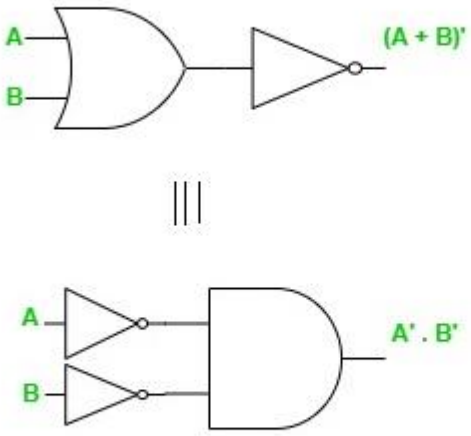
$$(A + B)' = A' . B'$$

Where

- *+ represents the OR operator between variables,*
- ***.** represents AND operator between variables, and*
- ***'** represents complement operation on variable.*

First De Morgan's Law Logic Gates

In context to logic gates and Boolean Algebra, De Morgan's Law states that "Both the logic gate circuits i.e., NOT gate is added to the output of OR gate, and NOT gate is added to the input of AND gate, are equivalent. These two logic gate circuits are given as follows:



First De Morgan's Law Truth Table

The truth table for first De Morgan's Law is given as follows:

A	B	A + B	(A + B)'	A'	B'	A'. B'
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

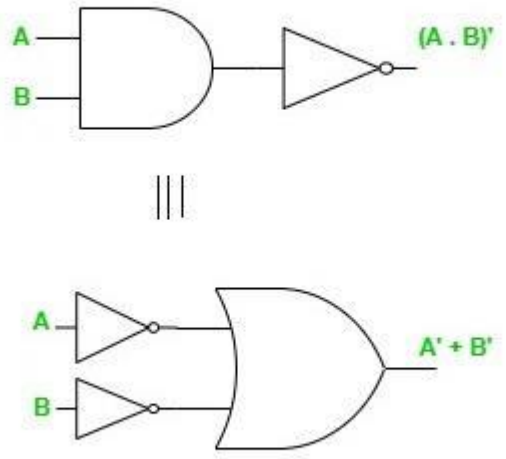
2 .Second De Morgan's law states that **"The complement of AND of two or more variables is equal to the OR of the complement of each variable."**
Let A and B be two variables, then mathematically Second De Morgan's Law is given as:

$$(A \cdot B)' = A' + B'$$

Second De Morgan's Law Truth Table

The truth table for the second De Morgan's Law is given as follows:

A	B	A . B	(A. B)'	A'	B'	A' + B'
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0



RESULT: Verified demorgans theorem successfully.